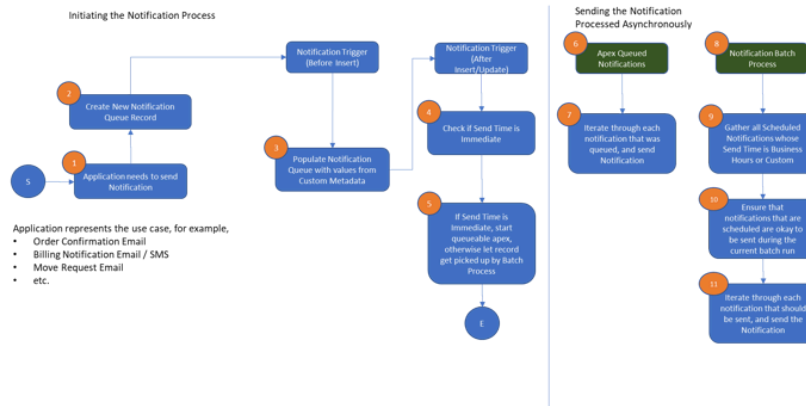


RUBIX - Notification Queue Documentation

This document contains an overview of how the Notification Queue process was designed and functions.

Overall Flow



1. The application is at a point where it needs to send a notification. Application is a generic term used to represent the use case. For example, when we need to send a Bill Notification to a customer, the application is responsible for determining whether a notification should be sent out, to who, whether it should be sent immediately or during business hours, and on what channel (email or SMS). Another example would be when we need to send an Order Confirmation email to the customer. The order submission process is responsible for determining whether the email should be sent out, and if so, who should it be sent out to, and whether it should be sent immediately or during business hours.

2. The application is responsible for creating the Notification_Queue__c record, and setting its attributes accordingly. There are two ways in which you can set the attributes on a Notification record.

- a. Using a Notification Queue Setting custom metadata record to prefill the Notification Queue record (see API Usage section for details on how to do this)
- b. Set the attributes on the Notification Queue record directly (see API Usage for more details)

Ultimately, once you've set the required attributes based how you want to prefill the notification queue record, the application must insert the Notification Queue record. Once inserted, the underlying Notification Queue process will take it from there.

3. If the application will use a Notification Queue Setting Custom Metadata record to prefill the Notification Queue record, then the Notification Queue's Before Insert trigger handler will look up the setting, and use the values on the custom metadata to prefill the notification queue record.

4. In the Notification Queue's After Insert / Update handler, the process will build a list of all Notification Queue records whose Send Time is set to Immediate

5. The process will then pass all of the records whose send time is set to Immediate to an Apex Queueable class (NotificationQueueable), and will then be enqueued for processing (using Apex Queues)

6. Apex Queued Notifications (Apex Class NotificationQueueable) is an asynchronous process that will process any Notification Queue records that are given to it. By running the notification process on a separate thread, this allows the application to move forward with any additional processes that are needed, and not worry about the delivery of the notification.
7. The asynchronous process calls the centralized Notification Queue processing function, NotificationQueueService.SendNotifications. The delivery of every notification, whether it's through queued apex, or through batch processing, goes through a standard process so that all emails or sms messages are sent in a consistent manner.
8. A batch process will be scheduled to run several times a day, retrieving all Notification Queue records that are eligible for delivery (see Notifications Batch Process for eligibility details)
9. All Notification Queue records whose Send Time is set to Business Hours or Custom are retrieved and made available for further analysis to determine whether the notification should be processed during the given batch cycle
10. Iterate through each Notification Queue record and determine whether the notification can be processed
 - a. If Send Time is set to Business Hours, the process will use the value stored on the Notification Queue's Business_Hours_Name__c attribute, and look up the Salesforce Business Hour setting that has the matching name. The process will then determine whether the batch cycle is currently running within the days / times configured on the Business Hour setting. If so, the process will send the Notification Queue record for processing
 - b. If Send Time is set to Custom, the process will check the date / time value stored on the Notification Queue's Custom_Send_Time__c attribute to see if the date/time has passed. If so, the process will send the Notification Queue record for processing
11. The batch process calls the centralized Notification Queue processing function, passing all notification records that are eligible for processing.

Object Model

Object / Field Name	Channel Used For (Email / SMS)	Data Type	Description
Notification_Queue__c			
Attachment_FileNames__c	Email	Long Text Area	Semi-colon (;) list of file names that correlate to the attachment URLs on the same record. These are the names of the files that will be attached in the email that is sent.

Attachment_URLs__c	Email	Long Text Area	URLs of any required Email attachments separated by a semi-colon (;). The underlying process will use the URL to retrieve the attachment that should be sent in a given email
	Attachment File Names and Attachment URLs are used together. If there are multiple attachments that must be sent in a given email, then the file name and urls must have the same sequence in the semi-colon delimited strings. For example: We need to include two attachments (Terms_Conditions.pdf, which can be downloaded from https://cdn/generalterms.pdf, AND BonusAttachment.pdf, which can be downloaded from https://cdn/bonus.pdf. The values of each attachment attribute would be as follows: Attachment_FileNames__c = Terms_Conditions.pdf;BonusAttachment.pdf Attachment_URLs__c = https://cdn/generalterms.pdf;https://cdn/bonus.pdf		
BCC_Email_Addresses__c	Email	Long Area Text	List of email addresses to BCC the Email Notification to, separated by a semi-colon (;)
Business_Hours_Name__c	Both	Text	Name of the Salesforce Business Hours configuration that the batch process should refer to determine whether the notification can be sent
CC_Email_Addresses__c	Email	Long Area Text	List of email addresses to CC the Email Notification to, separated by a semi-colon (;)
Custom_Send_Time__c	Both	Date / Time	Date & Time in which the notification should be sent. Once this date / time has been reached, the next batch run will process the notification queue record

Email_HTML_Body__c	Email	Rich Text Area (131,072)	Populate when sending an email whose content does not derive from a template. The body will only be used when a template is not specified.
Email_Subject__c	Email	Text	Populate when sending an email whose content does not derive from a template. The subject will only be used when a template is not specified.
Error_Details__c	Both	Long Area Text	In the event an error occurs, details of the error will be stored on this field for further troubleshooting
Org_Wide_Address_Name__c	Email	Text	Name of the Org Wide Email Address that should be used as the Sender when delivering an email. Required for all Email notifications
Related_Record_Id__c	Both	Text	ID of the related record that the notification gets merge field data from (the WhatId) For example, a billing invoice notification will have it's Related_Record_Id set to the CC Invoice record, or a Credit & Collection email will have it's Related_Record_Id set to the Credit and Collection record. NOTE: If the object being referenced as the Related Record does not Allow Activities, then the sent

			email will not show up on the target or related record's Activity Timeline.
Related_Record_Link__c	Both	Formula (Text)	A Hyperlink that a user can click on so that they can easily navigate to the related record
Reply_To_Email_Address__c	Email	Text	Email address that will be set as the "Reply To" on outgoing emails. If this field is blank, the default sender email address will be used as the reply to.
Results_Class_Name__c	Both	Text	In the event the “application” needs to know when a notification was processed, you can create a apex class that will “listen” for notifications being completed, and perform any extra processing as needed. The apex class must implement the interface NotificationQueueResultsInterface, and the Results_Class_Name__c must be set to the name of the apex class that will perform this extra processing. See API Usage for more details
Send_Time__c	Both	Picklist	Possible Values: • Immediate - Notification will be queued up and sent right away • Business Hours - Notification will be sent

			during the business hours specified on the Business_Hours_Name__c field Custom - Notification will be sent on or after the date / time specified on the Custom_Send_Time__c field
Setting_Name__c	Both	Text	Developer/API Name of the Notification Queue Setting Custom Metadata Type to get default values set from
SMS_Messaging_Channel_Name__c	SMS	Text	Developer Name of the Messaging Channel that should be used to send the SMS message i.e. Text_CA_13068051741
SMS_Phone_Number__c	SMS	Text	Phone Number that SMS Message will get sent to
Status__c	Both	Picklist	Possible Values - Scheduled - Notification is pending delivery - Completed - Notification has been delivered. The underlying process will automatically change the status to Completed upon successful delivery - Error - An error occurred while sending the notification - Manually Reprocessed - Indicates that this

			notification was manually sent Cancelled - Indicates the notification no longer needs to be sent
Target_Record_Id__c	Both	Text	ID of the Contact, Lead, or User record that the notification is getting sent to. (Who Id)
Target_Record_Link__c	Both	Formula (Text)	A Hyperlink that a user can click on so that they can easily navigate to the target record
Template_API_Name__c	Both	Text	API Name of the Email or SMS template that should be used to build the body of the notification being sent
To_Email_Addresses__c	Email	Long Text Area	List of email addresses to send the Email Notification to, separated by a semi-colon (;). Can leave this field blank if the application is attempting to send a notification to the Target Record.
Type__c	Both	Picklist	Possible Values - Email - Represents an Email Notification - SMS - Represents a SMS Notification

Notification_Queue_Setting__mdt

Custom Metadata record used to pre-fill a Notification Queue record) The purpose of this metadata is to allow for configurable changes to a specific notification without having to redeploy code. For example, in the event we want to change the template being used for a given notification, or whether a specific notification should go out during business hours instead of immediately, an administrator can change the values in the given custom metadata record, and all future notifications of that type will use the new values.

The pre-fill process will only populate the corresponding Notification_Queue__c record's field, when that field is empty. If the "application" has already populated a field on the Notification_Queue__c record, then the pre-fill

process will not attempt to update that field.

DeveloperName	Both	Text	Used to identify the custom metadata setting. The developer name follows the following naming convention: BusinessName_NotificationType_Channel_SendTime For Example: Energy_Bill_Notification_Email_BH - represents a notification for Energy, the notification is a Bill Notification, the notification is sent via Email, and the notification should be sent during business hours Energy_SMB_Order_Confirm_Email_I - represents a notification for Energy, the notification is a SMB Order Confirmation message, the notification is sent via Email and the notification should be sent immediately Send Times • I - Immediate • BH - Business Hours • C - Custom
Attachment_FileNames__c	Email	Text Area	Used to prefill the Notification_Queue record's Attachment_FileNames__c field.
Attachment_URLs__c	Email	Long Text Area	Used to prefill the Notification_Queue

			record's Attachment_URLs__c field
The values in the attachment fields follow the same convention as they do on the Notification_Queue__c object. File names and URLs must be in the same sequence.			
BCC_Email_Addresses__c	Email	Long Text Area	Used to prefill the Notification_Queue record's BCC_Email_Addresses__c field. Can be used to force all notifications to BCC an email address(es)
Business_Hours_Name__c	Both	Text	Used to prefill the Notification_Queue record's Business_Hours_Name__c field
CC_Email_Addresses__c	Email	Long Text Area	Used to prefill the Notification_Queue record's CC_Email_Addresses__c field. Can be used to force all notifications to CC an email address(es)
Custom_Send_Time__c	Both	Text - 5	Used to prefill the Notification_Queue record's Custom_Send_Time__c field. Value must be in the 24 hour format of HH:MM. When this setting is prefilled, if the time specified has already passed for the current day, then the Notification_Queue__c's Custom_Send_Time__c field will be set to tomorrow's date at the time specified, otherwise, the prefill process will use

			today's date and the time specified.
Email_Subject__c	Email	Text	Used to prefill the Notification_Queue record's Email_Subject__c field. Same rule applies, subject will only be used if template is not specified
Org_Wide_Address_Name__c	Email	Text	Used to prefill the Notification_Queue record's Org_Wide_Address_Name__c field
Reply_To_Email_Address__c	Email	Text	Used to prefill the Notification_Queue record's Reply_To_Email_Address__c field
Results_Class_Name__c	Both	Text	Used to prefill the Notification_Queue record's Results_Class_Name__c field
Send_Time__c	Both	Picklist	Used to prefill the Notification_Queue record's Send_Time__c field
SMS_Messaging_Channel_Name__c	SMS	Text	Used to prefill the Notification_Queue record's SMS_Messaging_Channel_Name__c field
Template_API_Name__c	Both	Text	Used to prefill the Notification_Queue record's Template_API_Name__c field
Type__c	Both	Picklist	Used to prefill the Notification_Queue

Notification Queue Setting - Custom Metadata Usage

The advantage of having Notification Queue Settings is that the details of a given notification can be stored in a single place, and if needed can be changed by an administrator.

As mentioned above, if the business decides to change when certain notifications are sent out, all it takes is a minor configuration change and the next notification to be created will automatically use the new notification setting.

The following Notification Queue Settings are currently in place.

Name	Description
Energy_Bill_Notification_Email_BH	Bill Invoice Notification sent via Email during Business Hours. This is used when a CC Invoice record is created, and the customer wants to receive an email notification that a new invoice is available. This is set to run during business hours so we do not sent invoice notifications in the middle of the night when CC invoices are being created in Salesforce.
Energy_Bill_Notification_SMS_BH	Bill Invoice Notification sent via SMS during Business Hours. This is used when a CC Invoice record is created, and the customer wants to receive a SMS notification that a new invoice is available. This is set to run during business hours so we do not sent invoice notifications in the middle of the night when CC invoices are being created in Salesforce.
Energy_Contract_Cancellation_Email_I	Contract Cancellation Email sent Immediately. This is used when a CSR checks the Send Cancellation Email checkbox on a Contract

Energy_Credit_Collection_Email_BH	Credit and Collection Email sent during Business Hours. This is used when a credit and collection notification is generated in Salesforce, an email notification will get sent out to the customer during regular business hours.
Energy_Move_Request_Email_I	Move Request Email sent immediately. This is used when a CSR submits a Move Request on the contract
Energy_Payment_Arrangement_Email_I	
Energy_Payment_Mthd_Confirmation_Email_I	
Energy_Res_Order_Confirm_Email_BH	Residential Order Confirmation Email that is sent out during business hours. This is used to sent confirmation emails for contracts generated in the Evergreen process. This allows the evergreen process to run overnight, and send notifications during business hours.
Energy_Res_Order_Confirm_Email_I	Residential Order Confirmation Email that is sent immediately. This is used when a CSR or customer signs up for a Residential Energy plan.
Energy_Res_Order_Confirm_No_Att_Email_BH	
Energy_Res_Order_Confirm_No_Att_Email_I	
Energy_SMB_Order_Confirm_Email_BH	SMB Order Confirmation Email that is sent out during business hours. This is used to sent confirmation emails for contracts generated in the Evergreen process. This allows the evergreen process to run

	overnight, and send notifications during business hours.
Energy_SMB_Order_Confirm_Email_I	SMB Order Confirmation Email that is sent immediately. This is used when a CSR or customer signs up for a Residential Energy plan.
Energy_SMB_Order_Confirm_No_Att_Email_BH	
Energy_SMB_Order_Confirm_No_Att_Email_I	

API Usage

To use the Notification Queue process, the following can be performed

Send Notification using Setting

To send a notification using a pre-defined custom metadata setting, you can do the following:

Apex

- Call NotificationQueueService.createNotificationRecordForSetting, passing the following
 - Related Record Id
 - Target Record Id
 - Developer Name of Notification Queue Setting Custom Metadata record
- This will return a Notification_Queue__c record
- Insert this record when ready

Flow

- Add a Create Record Activity, configured to create Notification_Record__c object
- Ensure the Setting_Name__c is set to the custom metadata record that should be used
- Set the Related_Record_Id__c accordingly
- Set the Target_Record_Id__c accordingly

Send Notification with using setting

A notification can be sent without using a custom metadata setting. However, when doing so, you'll need to populate the fields on the Notification_Queue__c record that normally would have been populated by the custom metadata setting.

Sending Email

To send and email, populate the Notification_Queue__c record with the following

With a Template

Field	Value
Type__c	Email
Send_Time__c	Immediate, Business Hours or Custom
To_Email_Addresses__c	Optional - Email address that should receive email
Org_Wide_Address_Name__c	Name of Org Wide Address to use
Template_API_Name__c	Developer Name of Email Template to use
Related_Record_Id__c	Id of record that template will use to generate content
Target_Record_Id__c	Id of Contact, Lead, User that email will be associated with. Must set if not setting To_Email_Addresses__c

Without a Template

Field	Value
Type__c	Email
Send_Time__c	Immediate, Business Hours or Custom
To_Email_Addresses__c	Optional - Email address that should receive email
Org_Wide_Address_Name__c	Name of Org Wide Address to use
Related_Record_Id__c	Id of record that email will be associated with
Target_Record_Id__c	Id of Contact, Lead, User that email will be associated with. Must set if not setting To_Email_Addresses__c
Email_Subject__c	Subject to be used in Email
Email_HTML_Body	Content that should make up the body of the email

Sending SMS

SMS should be sent using a Template

Field	Value
Type__c	SMS
Send_Time__c	Immediate, Business Hours or Custom
SMS_Phone_Number__c	Optional - Email address that should receive email
Org_Wide_Address_Name__c	Name of Org Wide Address to use
Template_API_Name__c	Developer Name of SMS Template to use
Related_Record_Id__c	Id of record that template will use to generate content
Target_Record_Id__c	Id of Contact, Lead, User that message will be associated with.

Send Notification Process

The purpose of the Notification Queue process is to have emails / sms sent out in a consistent manner. This ensures certain things like, error handling, email activities on objects, turning off signatures, and more, are done consistently across the board.

Email

- Gather IDs of all email templates and org wide email addresses referenced in Notification Queue records to be processed. This reduces the number of SOQL statements to execute
- Emails are sent out using the Messaging.SingleEmailMessage and Messaging.sendMail
- Iterate through each Notification Queue record
 - Setting the Body of the Email
 - If the Notification_Queue record has it's Template_API_Name__c populated, then use the template to populate the email body
 - Otherwise, use Notification_Queue__c.Email_Subject__c and Email_Html_Body__c to populate the subject and body of the email.
 - Set the Org Wide Email Address
 - If the Notification_Queue__c.Reply_To_Email_Address__c is populated, then set it accordingly on the Messaging.SingleEmailMessage object by calling setReplyTo
 - If the Notification_Queue__c.To_Email_Addresses__c is populated, then convert the semi-colon delimited string to a list, and pass it to the Messaging.SingleEmailMessage object by calling setToAddresses. When this occurs, the setTreatTargetObjectAsRecipient is also set to false
 - If the Notification_Queue__c.CC_Email_Addresses__c is populated, then convert the semi-colon delimited string to a list, and pass it to the Messaging.SingleEmailMessage object by calling setCcAddresses

- If the Notification_Queue__c.BCC_Email_Addresses__c is populated, then convert the semi-colon delimited string to a list, and pass it to the Messaging.SingleEmailMessage object by calling setBccAddresses
- If the Notification_Queue__c.Attachment_URLs__c and Attachment_FileNames__c is populated, then convert the semi-colon delimited string to a list, and iterate through each instance
 - Retrieve the content specified in the URL
 - Create an instance of Messaging.EmailFileAttachment
 - set the fileName using setFileName
 - set the Body using SetBody
 - Add all attachments for the given notification using Messaging.SingleEmailMessage.setFileAttachments
- If the Notification_Queue__c.Target_Record_Id__c is not blank then pass it to the Messaging.SingleEmailMessage object by calling setTargetObjectId
- If the Notification_Queue__c.Related_Record_Id__c is not blank then pass it to the Messaging.SingleEmailMessage object by calling setWhatId
 - If both the related object and target object have Allow Activities turned on at the SObject level, then the Messaging.SingleEmailMessage.setSaveAsActivity will be set to True, otherwise, it will be set to False
- Set Messaging.SingleEmailMessage.setUseSignature to false
- Send the email messages using Messaging.sendEmail
 - Emails are sent using an indicator that attempts to send all emails, regardless if any of them fail
- Analyze the result
 - The process will iterate through all instances of the Messaging.SendEmailResult list that is returned by Messaging.sendEmail
 - If isSuccess is set to True
 - The Notification_Queue__c.Status__c is set to Success
 - Otherwise
 - The Notification_Queue__c.Status__c is set to Error
 - The Notification_Queue__c.Error_Details__c is set to the details stored on the Messaging.SendEmailError instance

SMS

- Iterate through all SMS Notification Queue records to be processed
 - Invoke the Energy_Send_SMS_Notification flow, passing
 - MessagingChannelName = Notification_Queue__c.SMS_Messaging_Channel_Name__c
 - MessagingTemplateName = Notification_Queue__c.Template_API_Name__c
 - RelatedRecordId = Notification_Queue__c.Related_Record_Id__c
 - RecipientPhoneNumber = Notification_Queue__c.SMS_Phone_Number__c
 - RecipientRecordId = Notification_Queue__c.Target_Record_Id__c
 - The flow will use the Send SMS activity to send the actual SMS message to the customer

- Analyzing the Result
 - If the output variable “Result” is not blank, then we assume the delivery was a success, and the Notification_Queue__c.Status set to ‘Completed’
 - Otherwise, If the output variable “Result” is blank, then we assume a delivery error, and the Notification_Queue__c.Status__c is set to ‘Error’

After all Notification_Queue__c records have been processed, the records are updated so that their updated Status is saved to the database.

A final iteration is performed is then performed on the records, calling any Notification Queue callback functions that have been configured on the Notification_Queue__c record(s), passing success, failures, and skipped.

Apex Queued Process

When calling NotificationQueueService.QueueNotifications, the process will iterate through each instanced passed, and make a list of all records whose the Send_Time__c is set to “Immediate”

This list is then passed to a new instance of NotificationQueueable, and this NotificationQueueable instance is passed into System.enqueueJob

The queueable job itself calls NotificationQueueService.SendNotifications, passing all Notification_Queue__c records that were initially provided to it. This the method that is described above.

Notifications Batch Process

The batch process is schedule to run 7 days a week at 30 minute intervals at batches of 50 records. When it executes, it will do the following:

- Retrieve all notification queue records where
 - Status__c is set to Scheduled
 - AND
 - Send_Time__c = “Custom” and Custom_Send_Time__c <= Now
 - OR Send_Time__c = ‘Business Hours’
- Iterate through all retrieved records
 - If Send_Time__c = “Custom”
 - Add record to a list of notifications that will be processed
 - If Send_Time__c = “Business Hours”
 - Get the Salesforce Business Hours configuration that corresponds to the Business_Hours_Name__c on the Notification_Queue__c record
 - Check to see if the current time is within the Business Hours configured, if so, then add the Notification_Queue__c record to a list of notifications that will be processed, otherwise, skip it

- Pass all eligible notification records to the NotificationQueueService.SendNotifications function

Business Hours

To configure a notification to be sent out during business hours configured within Salesforce, ensure the following has been set on the Notification_Queue__c record, either by pre-filling with a corresponding Notification_Queue_Setting__mdt record, or setting it programmatically.

- Send_Time__c = "Business Hours"
- Business_Hours_Name__c = Name of Salesforce Business Hour configuration

Ensure you have Business Hour configured in the org, and that the name is matching what is on the Notification_Queue__c record

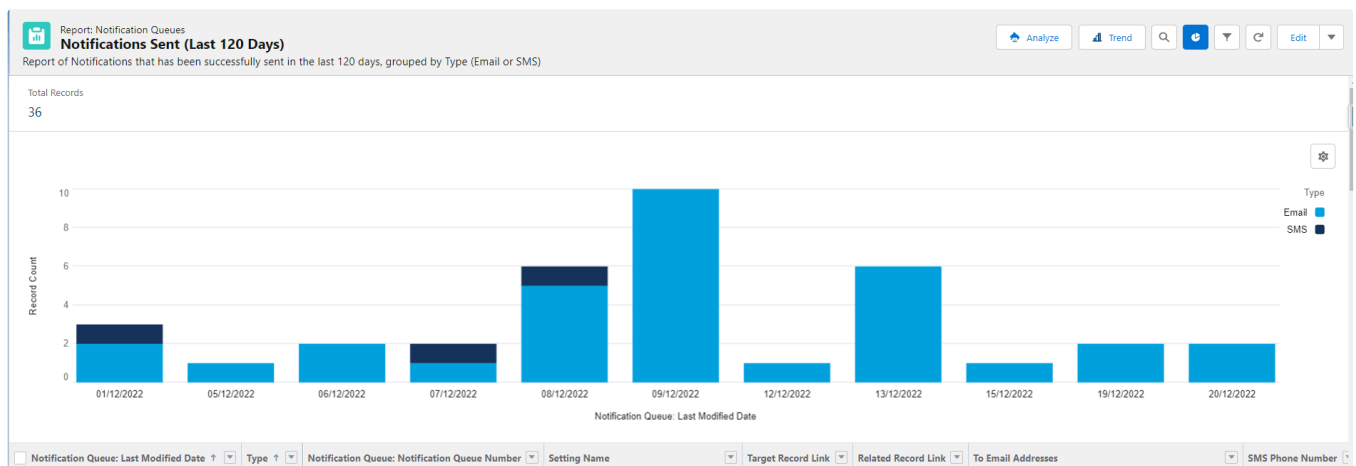
Reports

The following reports are available for tracking Notifications. They are all located under RUBIX Reports > Notification Reports

- Notifications Error Report - Shows Notifications that have resulted in an error.

Report: Notification Queues Notifications Error Report								
Contains a list email / sms notifications that experienced an error when attempting to send the notification.								
Total Records 3								
	Notification Que--	Notification--	Error Details	Target Record Link	Related Record Link	Send Time	Type	Setting Name
1	NOTI-000035	15/12/2022	Error Details: SaveAsActivity is not allowed with whatid that is not supported as a task whatid.	Recipient	Related Record	Immediate	Email	Energy_Move_Request_Email_I
2	NOTI-000025	15/12/2022	Check the Messaging Templates Error Log for more information.	Recipient	Related Record	Business Hours	SMS	Energy_Bill_Notification_SMS_BH
3	NOTI-000034	09/12/2022	Error Details: The email template is not active and cannot be used.	Recipient	Related Record	Immediate	Email	Energy_Move_Request_Email_I

- Notifications Sent (Last 120 Days) - Shows all Email and SMS notifications that have been successfully sent in the last 120 days. Details are in the table below the chart



- Notifications Set by Message Type - Shows all Notifications successfully sent, grouped by their Setting_Name__c. The Setting Name typically indicates which Notification_Queue_Setting__mdt record was

used to prefill the Notification_Queue__c record.



- Notifications to be Sent - Shows all notifications that are pending processing (i.e. Status = Scheduled). This can give an administrator an idea of how many notifications will go out in the next batch run.

